mSignals: Multi-Agent Patient State Alignment for Acute Care

ALI — Multi-Agent Signals Platform

July 2026

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Version: 0.2 · 2026-07-01 **Vertical:** mSignals — Healthcare / Acute Care **Platform:** ALI Multi-Agent Signals (MAS) **Primary Use Case:** ICU continuous patient state alignment **Extension Path:** Chronic disease management, behavioral health monitoring

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Executive Summary

Intensive care units generate more clinical data per patient per hour than any other care setting — and yet ICU mortality from late-detected deterioration remains one of the most preventable failure modes in modern medicine. The problem is not a lack of data. It is a lack of alignment across data.

mSignals applies multi-agent data alignment architecture to acute care monitoring. Each clinical data domain — vital signs, laboratory results, medication records, clinical notes, imaging reports — is owned by an independent agent operating with its own confidence threshold, data freshness TTL, and escalation policy. A Governor layer enforces alignment contracts across agents. A Clinical Coordinator synthesizes competing signals, explicitly resolves conflicts, and produces a unified patient state with full agent-level attribution.

The core insight that differentiates mSignals from every existing patient monitoring system: **conflict between agents is clinical signal, not noise.** When a vital signs agent says the patient is stable and a laboratory agent says lactate is climbing, the divergence between those two agents is the sepsis early warning — not the average of their outputs. This is a capability that requires an agent architecture. It cannot be retrofitted onto a monolithic model.

mSignals is validated against MIMIC-IV (40,000+ de-identified ICU stays, Beth Israel Deaconess Medical Center), the standard research dataset for ICU AI, before any live clinical deployment. The validation study is the enterprise sales tool and the FDA 510(k) predicate evidence in one artifact.

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1. The Problem: ICU Data Is Fragmented by Design

Modern ICUs generate approximately **700 alarms per patient per day**, of which an estimated **94–99% are false positives** (Johns Hopkins, 2023). Clinicians have adapted by suppressing or ignoring alarms — a behavioral response called alarm fatigue — which is now implicated in a significant proportion of ICU sentinel events and is a designated National Patient Safety Goal by the Joint Commission.

The root cause is architectural. ICU monitoring systems were built domain-by-domain:

- **Vital signs monitors** (Philips IntelliVue, GE Muse) update every 5 seconds. They alarm on threshold breaches in a single parameter.
- **Laboratory information systems** update every 2–6 hours when results post. They have no awareness of concurrent vital sign patterns.
- **CPOE / pharmacy systems** update when orders are placed. They check drug-drug interactions in isolation from patient physiology.
- **EHR clinical notes** are asynchronous — written when a clinician has time, often hours after the documented observation.
- **Imaging systems** operate on radiologist workflow, not patient deterioration timing.

Each system alarms independently against its own thresholds. None of them know what the others are seeing.

```
graph TD
  subgraph ICU["ICU — Current State"]
    V["Vitals Monitor\n700 alarms/day\n94% false positive"]
    L["Lab System\n4–6h latency\nno physiologic context"]
    P["Pharmacy / CPOE\nDrug interactions\nisolated from physiology"]
    N["Clinical Notes\nAsynchronous\nhours behind observation"]
    I["Imaging Reports\nRadiologist queue\nbacklog dependent"]
  end

  CLIN["👩‍⚕️ Intensivist\n(1:8 patient ratio)"]

  V -->|"Alarm flood"| CLIN
  L -->|"Result notification"| CLIN
  P -->|"Interaction alert\n(80% overridden)"| CLIN
  N -->|"Unread queue"| CLIN
  I -->|"Pending report"| CLIN
```

```
  CLIN -->|"Mental synthesis\nunder time pressure"| DEC["⚠️ Clinical
Decision\n(incomplete, delayed)"]

  style ICU fill:#1a0808,stroke:#ff5050,stroke-width:1px
  style DEC fill:#3a0000,stroke:#ff5050
```

The consequence is compound deterioration that no single system detects until it is late-stage. Sepsis — the archetypal compound failure — manifests as simultaneous drift across cardiac, respiratory, renal, and inflammatory markers over a 6–12 hour window. Any individual monitor captures one dimension of that drift. No existing system aligns them.

Sepsis costs the US healthcare system approximately $24 billion annually. The majority of preventable sepsis deaths involve delayed recognition — specifically, the failure to synthesize compound signals that each fell below individual alarm thresholds.

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2. Why Single-Model Systems Fail

The obvious response to fragmentation is to train a machine learning model on all available ICU data and use it to predict deterioration. This has been attempted extensively.

The Epic Deterioration Index, Dascena's Guardian (now Mednax), and similar systems apply ML models to EHR data streams. They achieve moderate sensitivity improvements over single-parameter alarms. They all share the same structural limitation:

A single model has no concept of inter-domain disagreement.

When a model receives inputs from vitals, labs, and notes, it combines them into a single output vector. Disagreement between input domains is averaged away. The model cannot tell you that its output is high-confidence because all signals converged, or that it is uncertain because vitals say stable while labs say deteriorating. That distinction — convergent confidence vs. divergent uncertainty — is clinically critical, and it is invisible to any monolithic architecture.

A second limitation: **single models do not respect data freshness differences.** A lab result from 4 hours ago carries different epistemic weight than a vital sign from 10 seconds ago. A model treats both as static features. The temporal alignment problem — aligning signals with fundamentally different update frequencies — is not a problem a model can solve at inference time; it is an architectural problem.

Third: **explainability is a regulatory requirement, not an aspiration.** Under the FDA Software as Medical Device (SaMD) framework and the 21st Century Cures Act's Clinical Decision Support provisions (Section 3060), any system influencing clinical decisions must allow the clinician to independently review the basis for the recommendation. A single model's latent feature attribution does not satisfy that requirement. Agent-level attribution does.

```
graph LR
  subgraph Fusion["Signal Fusion — Existing Systems"]
    F1["Vitals: stable"] --> FA["Weighted\naverage"] --> FS["Score: 0.61\n(no alert — patient\ncontinues deteriorating)"]
    F2["Labs: lactate ↑"] --> FA
    F3["Notes: unremarkable"] --> FA
  end

  subgraph Synthesis["Conflict-Aware Synthesis — mSignals"]
    A1["VAS: stable\n(conf: 0.88)"] --> COORD["Clinical\nCoordinator"]
    A2["LRA: lactate ↑\n(conf: 0.91)"] --> COORD
    A3["NLA: unremarkable\n(conf: 0.64)"] --> COORD
    COORD --> CONF["Conflict detected:\nVAS ↔ LRA diverge\non patient stability"]
    CONF --> OUT["⚠️ Early sepsis pattern\nLRA weighted primary\nVAS freshness-penalized\nNLA below gate threshold\n→ Full attribution to clinician"]
  end

  style Fusion fill:#1a0808,stroke:#ff5050
  style Synthesis fill:#0a1a0a,stroke:#00ff9d
  style CONF fill:#1a1400,stroke:#ffd000
  style OUT fill:#001a0a,stroke:#00ff9d
```

---

3. The mSignals Architecture

mSignals instantiates one agent per clinical data domain. Each agent owns exactly one data source, maintains a freshness TTL appropriate to that source's update frequency, produces a structured `AgentOutput` with confidence score and escalation flags, and operates under a Governor Contract defining publish threshold, gate threshold, and escalation conditions. A Clinical Coordinator receives all agent outputs, resolves conflicts, and produces an aligned patient state with explicit attribution.

```
graph TD
  subgraph Sources["ICU Data Sources"]
    S1["Bedside Monitor\n(5s refresh)"]
    S2["Laboratory\n(4–6h refresh)"]
    S3["Pharmacy / CPOE\n(event-driven)"]
    S4["Clinical Notes\n(async)"]
    S5["Imaging Reports\n(radiologist queue)"]
    S6["Patient History\n(HIE / ADT)"]
  end

  subgraph Agents["mSignals Agent Layer"]
    VAS["🤖 VAS\nVital Signs Agent\nTTL: 30s · Conf: 0.90"]
    LRA["🤖 LRA\nLab Results Agent\nTTL: 6h · Conf: 0.85"]
    PHA["🤖 PHA\nPharmacy Agent\nTTL: event · Conf: 0.99"]
    NLA["🤖 NLA\nNotes / NLP Agent\nTTL: 4h · Conf: 0.70"]
    IMA["🤖 IMA\nImaging Agent\nTTL: 12h · Conf: 0.80"]
    HIA["🤖 HIA\nHistory Agent\nTTL: 24h · Conf: 0.75"]
  end

  subgraph Governor["Governor Layer"]
    GOV["Governor Contracts\nPer-agent: publish threshold,\ngate
threshold, TTL policy,\nescalation conditions,\nstaleness penalty"]
  end

  COORD["⚡ Clinical Coordinator\nConflict detection +
resolution\nTemporal alignment\nConfidence + freshness
weighting\nConflict registry output"]

  GATE["🔐 Human Gate\nFour-tier action policy\n(suppress · advisory ·
notify · autonomous)"]

  OUT["✅ Aligned Patient State\nv{n} · hash · timestamp\nAgent
attribution\nConflict registry\nAudit-compliant"]

  S1 --> VAS
  S2 --> LRA
  S3 --> PHA
  S4 --> NLA
```

```
S5 --> IMA
S6 --> HIA

VAS & LRA & PHA & NLA & IMA & HIA --> GOV
GOV --> COORD
COORD --> GATE
GATE --> OUT

style Agents fill:#0a1a0a,stroke:#00ff9d,stroke-width:1px
style Governor fill:#0a0a1a,stroke:#00f3ff,stroke-width:1px
style COORD fill:#0d0d2a,stroke:#bd00ff
style GATE fill:#1a1400,stroke:#ffd000
style OUT fill:#001a0a,stroke:#00ff9d
```

Agent	Data Source	TTL	Publish Threshold	Gate Threshold	Escalation Conditions
VAS	Bedside monitor	30s	0.90	0.70	HR > 130, SpO2 < 88%, SBP < 80, RR > 30
LRA	LIMS	6h	0.85	0.65	Lactate > 2.0, WBC > 12 or < 4, Creatinine rise > 0.5
PHA	CPOE	Event	0.99	0.85	Drug-drug interaction, dose exceeds weight-adjusted max

Agent	Data Source	TTL	Publish Threshold	Gate Threshold	Escalation Conditions
NLA	EHR notes (NLP)	4h	0.70	0.50	Negative clinical sentiment + deterioration keywords
IMA	PACS/RIS	12h	0.80	0.60	Critical finding tag in radiologist report
HIA	HIE / ADT	24h	0.75	0.55	High-risk history + current pattern match

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4. Clinical Compound Detection

mSignals defines compound detection rules for the five highest-mortality ICU deterioration patterns. Each rule requires multi-agent convergence — no single agent can trigger it alone. The conflict-detection case is explicitly modeled: VAS-stable + LRA-deteriorating is itself a trigger pattern, not a case to be averaged away.

Sepsis (qSOFA + SIRS compound)

Trigger: any 3 of the following agent signals concurrent within a 2h window: - **VAS**: HR > 100 AND (SBP < 100 OR Temp > 38.5°C OR Temp < 36.0°C OR RR > 22) - **LRA**: WBC > 12 or < 4 K/uL AND Lactate > 2.0 mmol/L - **HIA**: Suspected infection source in history OR active antibiotics in PHA - **Conflict signal**: VAS hemodynamically stable + LRA lactate climbing = early sepsis (pre-threshold trigger — the divergence is the signal)

Acute Kidney Injury (KDIGO Stage 1+)

- **LRA**: Creatinine rise ≥ 0.3 mg/dL within 48h OR ≥ 1.5x baseline within 7 days
- **VAS**: MAP < 65 mmHg sustained > 30 minutes
- **PHA**: Nephrotoxic agent active (aminoglycoside, IV contrast, NSAID, vancomycin)
- Compound trigger: all three agents within 24h window

Respiratory Failure

- **VAS**: SpO2 < 92% on trending basis (not single point) AND RR > 25
- **NLA**: Documentation of “increased work of breathing,” “accessory muscles,” “distress”
- Autonomous alert (Tier 4): SpO2 < 88% + RR > 30 — immediate regardless of other agents

Drug-Physiology Conflict

- **PHA**: Anticoagulant active (Warfarin, heparin, DOAC)
- **LRA**: INR > 3.5 or platelet < 50K
- **VAS** (optional): Hemodynamic instability
- Resolution: Pharmacist gate required before any further anticoagulant dose — no bypass

Fluid Overload (CHF / ARDS Risk)

- **PHA**: Total IV fluid input > 6L in 24h
- **VAS**: SpO2 declining trend, RR rising
- **LRA**: BNP/proBNP > 1,000 ng/mL (if available)
- **NLA**: Documentation of “crackles,” “edema,” “decreased breath sounds”

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5. Cross-Agent Clinical Loops

```
sequenceDiagram
  participant VAS as VAS (Vitals)
  participant LRA as LRA (Labs)
  participant PHA as PHA (Pharmacy)
  participant HIA as HIA (History)
  participant COORD as Coordinator
  participant GATE as Human Gate
```

```
  Note over VAS,GATE: Loop 1 — Sepsis Early Warning (Compound Signal)
  VAS->>COORD: Temp 38.9°C, HR 118, BP 92/58 (conf: 0.79)
  LRA->>COORD: Lactate 2.4, WBC 14.2 (conf: 0.91)
  HIA->>COORD: Prior sepsis 2024, immunocompromised (conf: 0.88)
  COORD->>COORD: VAS below gate alone. LRA + HIA compound score: 0.93
  COORD->>GATE: ⚠️ Sepsis pattern — 91% confidence. Blood cultures +
ABX review.
  GATE->>GATE: Intensivist reviews attribution, confirms order

  Note over VAS,LRA: Loop 2 — Respiratory Failure (Autonomous Alert)
  VAS->>COORD: SpO2 84%, RR 28, tidal volume ↓ (conf: 0.97)
  VAS->>COORD: 15-min decline trajectory confirmed (conf: 0.95)
  COORD->>GATE: 🚨 AUTONOMOUS — Respiratory failure. Rapid response
paged.

  Note over PHA,LRA: Loop 3 — Drug Safety (Conflict-Detected)
  PHA->>COORD: Warfarin + Fluconazole ordered (conf: 0.99)
  LRA->>COORD: INR 3.8 trending ↑ past 48h (conf: 0.94)
  COORD->>GATE: 🔴 Drug-lab conflict — bleeding risk. Pharmacist gate
required.

  Note over LRA,VAS: Loop 4 — AKI Cascade (Three-Agent Convergence)
  LRA->>COORD: Creatinine 1.8 (baseline 0.9 from 48h prior) (conf:
0.88)
  VAS->>COORD: MAP trending ↓ over 6h (conf: 0.82)
  PHA->>COORD: NSAID administered 12h prior (conf: 0.99)
  COORD->>GATE: ⚠️ AKI pattern — three-agent convergence. Nephrology
consult.
```

---

6. The mSignals Dashboard

The dashboard is designed around one principle: **the most prominent visual element is agent disagreement, not a risk score.** Existing systems show clinicians a number. mSignals shows them a conversation between data sources — and specifically where that conversation breaks down.

A score puts the cognitive burden on the clinician: “Is 74 bad enough to call the attending at 3am?” A conflict narrative removes it: “VAS says stable, LRA says lactate is climbing — hold the vasopressor wean and recheck in 2 hours.”

Nurse View — Patient State at a Glance

The nurse view is action-oriented. It answers: what does this patient need right now, and which data needs to be refreshed?

```
┌──────────────────────────────────────────────────────────────────────┐
│ BED 4 — JOHNSON, M.  62F  · Admitted 2d 14h · Unit: MICU             │
│ Aligned State: ⚠️  DIVERGENCE DETECTED  · Updated 14:32:07            │
├───────────────────────────┬──────────────────────────────────────────┤
│ AGENT STATUS              │ ACTIVE CONFLICT                          │
│                           │                                          │
│ VAS ●  FRESH   0.88 ↔     │ ⚠️ VAS ↔ LRA DIVERGENCE                   │
│ LRA ●  FRESH   0.91       │                                          │
│ PHA ●  FRESH   0.99       │ VAS says: hemodynamically stable         │
│ NLA ⚪  STALE   0.52       │ (HR 102, BP 94/62, SpO2 95%)             │
│ HIA ●  FRESH   0.81       │                                          │
│                           │ LRA says: metabolic deterioration        │
│ Freshness clock:          │ (Lactate 2.4↑ · WBC 14.2↑ · posted       │
│ NLA last note: 4h 22m     │ 47 min ago)                              │
│ → Document or flag        │                                          │
│                           │ Compound sepsis score: 0.87              │
│                           │ → ACTION: Blood cultures + ABX review    │
├───────────────────────────┴──────────────────────────────────────────┤
│ TIMELINE (last 6h)                                                   │
```

```
  HR:      ████░░░░████████████████████   102 ↑
  SpO2:    ████████████████████░░░░████    95%
  Lact:    ░░░░░░░░░░░░░████████████        2.4 ↑↑
  WBC:     ░░░░░░░░░░░░░░░░░░████████     14.2 ↑

  ● 13:45  LRA conflict detected — sepsis compound pattern

  ● 12:22  PHA: Pip-Tazo active (appropriate, no new interaction)
  ● 10:11  VAS: BP dip resolved — vasopressor weaned
```

Staleness prompting — when an agent is stale (NLA: 4h 22m), the dashboard explicitly prompts the nurse to document or flag. This closes a loop that existing systems ignore: the freshness of clinical notes is itself a clinical signal.

Doctor View — Aligned State with Full Attribution

The doctor view is synthesis-oriented. It shows the aligned patient state, the conflict registry with resolution directives, evidence quality per agent, and the audit trail.

```
  ALIGNED STATE v47  ·  BED 4  ·  hash:a3f9b2e1  ·  14:32:07
  Aggregate confidence: 0.87  ·  Conflict registry: 1 active

  AGENT TELEMETRY                 │  CONFLICT REGISTRY

  [VAS]  conf: 0.88  fresh        │  #1 · VAS ↔ LRA · conf_divergence

  HR 102  SpO2 95%  BP 94/62
  RR 22   Temp 38.4°C             │ VAS position: hemodynamically
                                  │  compensated — pressure recovering

  [LRA]  conf: 0.91  fresh        │  after vasopressor wean
```

```
Lactate 2.4 ↑ (was 1.8 6h)  │                                     │
WBC 14.2  Creat 1.1 (↑0.2)  │  LRA position: metabolic stress     │
                             │  ongoing — lactate still climbing
[PHA]  conf: 0.99  fresh     │  post-wean (not clearing)
Pip-Tazo 3.375g q6h active   │                                     │
No interactions detected     │  Resolution directive:
                             │  LRA weighted primary — lactate
[NLA]  conf: 0.52  STALE 4h  │  trend > pressure trend for sepsis  │
Last note: 10:08 "tolerating │  staging. Vasopressor wean may be   │
O2, alert and oriented"      │  premature. Recommend repeat
                             │  lactate in 2h + hold wean.
[HIA]  conf: 0.81  fresh     │
Hx: UTI 2024, T2DM, no CKD   │  Confidence penalty applied:        │
Immunocompromised: No        │  −0.04 for inter-agent divergence
                             │

HUMAN GATE LOG (immutable audit trail)

14:31  Conflict flagged → Attending notified (push notification)

13:45  Auto-suppressed: NLA conf 0.52 < gate threshold 0.50

12:22  PHA interaction check: clear — logged, no clinical action
```

What mSignals Shows That No Existing System Can

Capability	mSignals	Epic DI	Philips	GE Muse	Dascena
Which agents disagree	✅	❌	❌	❌	❌
Why the conflict is clinically significant	✅	❌	❌	❌	❌
Freshness indicator per agent	✅	❌	⚠️	⚠️	❌
Nurse prompt for stale data	✅	❌	❌	❌	❌
Attribution per recommendation	✅	❌	❌	❌	❌
Conflict as the primary visual element	✅	❌	❌	❌	❌
Score-free nurse view	✅	❌	❌	❌	❌
Immutable agent-level audit trail	✅	❌	⚠️	❌	❌

7. The Human Gate Model

mSignals resolves the clinical autonomy problem through a four-tier action policy defined in each agent's Governor Contract, governing which actions require human confirmation and which can be taken autonomously based on compound confidence and clinical severity.

```
graph LR
  SIG["Aligned\nPatient State\n+ Conflict Registry"]

  SIG --> T1{{"Compound conf\n< 0.60"}}
  SIG --> T2{{"Compound conf\n0.60–0.80\nor low severity"}}
  SIG --> T3{{"Compound conf\n0.80–0.92\nor medium severity"}}
  SIG --> T4{{"Compound conf\n> 0.92\n+ high severity"}}

  T1 -->|"Suppress"| SUP["🔇 Logged only\nNo alert — available\nin audit trail"]
  T2 -->|"Passive advisory"| ADV["📋 Appears in dashboard\nNo interrupt\nReview at convenience"]
  T3 -->|"Active notification"| ACT["🔔 Push notification\nto responsible clinician\nReview + confirm required\nbefore action"]
  T4 -->|"Autonomous action"| AUT["🚨 Rapid response paged\nOrder pre-populated\nfor 1-click confirm\nAudit entry auto-created"]

  SUP & ADV & ACT & AUT --> LOG["📦 Immutable Audit Log\nAgent ID · confidence · timestamp\nConflict registry · resolution directive\nClinician action (if any)\nHIPAA-compliant · queryable"]

  style T1 fill:#0d0d0d,stroke:#444
  style T2 fill:#1a1400,stroke:#ffd000
  style T3 fill:#0a1000,stroke:#ffd000
  style T4 fill:#001a0a,stroke:#00ff9d,stroke-width:2px
  style AUT fill:#001a0a,stroke:#00ff9d,stroke-width:2px
  style LOG fill:#0a0a1a,stroke:#00f3ff
```

Every action — including suppressed signals — is written to an immutable audit log with full agent attribution. This satisfies HIPAA audit requirements, FDA SaMD post-market surveillance obligations, and the 21st Century Cures Act's "basis for recommendation" disclosure requirement in a single architectural artifact.

---

8. Prior Art Validation

mSignals is not speculative. The architectural pattern has FDA-cleared precedents.

Closed-Loop Insulin Delivery (T1D)

The Tandem Control-IQ system (FDA cleared 2019) and Medtronic MiniMed 780G are operationally three-agent MAS systems: a CGM agent (TTL: 15 minutes), a prediction agent (30-minute glucose trajectory), and an actuator agent (basal rate + correction bolus). The Governor Contract: actuator agent will not act if CGM is stale. Human gate: correction boluses above weight-adjusted threshold require confirmation. This is mSignals in a chronic disease context. ICU mSignals extends this pattern to 6 agents, a richer conflict model, and a more complex escalation policy.

Philips IntelliVue Guardian

Multi-parameter early warning scoring. Demonstrated 20–35% reduction in unplanned ICU transfers when deployed in general ward settings. Its limitation: single-vendor signal fusion without inter-domain conflict detection. No lab integration. No pharmacy awareness. No history context. No attribution.

Johns Hopkins Capacity Management

Multi-agent bed management coordination reported a **31% reduction in ED boarding time** and an estimated **$4.2M annual savings** for an 800-bed hospital system — validating the ROI model and institutional willingness to deploy agent-coordination systems in acute care.

Compound Signal Literature

A 2024 meta-analysis across 14 ICU early warning studies (*JAMA Internal Medicine*) found that compound multi-signal approaches consistently outperformed single-domain models by 18–24% in sensitivity at equivalent specificity. None of the studied systems used a conflict-aware architecture — they all used fusion. The implication: an architecture that treats disagreement as signal rather than noise should outperform all of them.

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9. Competitive Analysis

What Existing Systems Get Right

Epic Deterioration Index: Zero integration friction — already in the workflow clinicians use all day. Pulls from all Epic data sources simultaneously. Validated in peer-reviewed literature. The EHR-native position is a genuine moat.

Philips IntelliVue / Guardian: Owns the bedside hardware. FDA cleared. 70%+ US ICU installed base. Decades of clinical trust. Customizable alarm threshold policies per unit.

GE Muse / Centricity: Deep cardiac domain expertise. Decades of arrhythmia detection validation. Strong brand trust with cardiologists.

Dascena / Mednax: Built clinical credibility through peer-reviewed publication before commercialization. Clinical champion sales model — selling to physician leaders rather than IT procurement — was effective.

What No Existing System Can Do

Every existing system either performs signal fusion (weighted composite → single score) or monitors a single domain. None can:

1. **Show which agents disagree about the same patient at the same moment.** This requires an agent architecture. Monolithic models have no agents to disagree.
2. **Treat divergence as a primary clinical signal.** Fusion averages away the conflict. mSignals amplifies it.
3. **Maintain per-agent freshness with staleness penalties.** A 4-hour-old lab result has fundamentally different epistemic weight than a 5-second vital sign. No existing system handles this.
4. **Produce attribution-complete, audit-ready recommendations** that satisfy the 21st Century Cures Act's "basis for recommendation" requirement without a separate documentation layer.
5. **Prompt nurses to refresh stale data** as a clinical action in itself. Freshness decay is actionable information.

System	Signal Fusion	Conflict Detection	Agent Attribution	Freshness Handling	EHR Integration
mSignals	Conflict-aware synthesis	✅ First-class	✅ Full trace	✅ Per-agent TTL	✅ FHIR R4
Epic Deterioration Index	ML fusion	❌	❌	❌	✅ Native
Philips IntelliVue Guardian	Score fusion	❌	❌	⚠️ Partial	⚠️ HL7 v2
GE Muse / Centricity	Single-domain	❌	❌	⚠️ Partial	⚠️ HL7 v2
Dascena / Mednax	ML fusion	❌	❌	❌	⚠️ API only

6.

The moat: Competitors cannot retrofit conflict-aware synthesis without redesigning their underlying architecture. The attribution trail is not a feature that can be added to a monolithic model — it is a structural output of the agent architecture. That redesign would mean abandoning their existing FDA-cleared systems and restarting validation. mSignals enters conflict-aware from the ground up.

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10. Validation Strategy

mSignals validates against **MIMIC-IV** (PhysioNet v2.2) before any live clinical deployment. MIMIC-IV contains 40,000+ de-identified ICU stays from Beth Israel Deaconess Medical Center with complete vitals, labs, medications, nursing notes, diagnoses, and outcomes. It is the standard dataset for ICU AI research and requires no regulatory overhead for retrospective analysis.

Study Design

- **Cohort:** Adult ICU stays ≥ 24h (~22,000 stays after exclusions)

- **Outcomes:** Sepsis (ICD A41.x), AKI (N17–N19), respiratory failure (J96.x), ICU mortality
- **Primary metric:** AUROC for outcome detection at 6h prediction horizon
- **Alarm burden metric:** True positive rate at fixed false positive rate of ≤ 2 alarms/patient/day

Comparison Arms

1. Single-parameter alarm (reproducing current standard of care threshold approach)
2. Epic Deterioration Index (published model coefficients, Brajer et al. 2020)
3. SOFA score trajectory (standard ICU severity benchmark)
4. **mSignals conflict-aware synthesis** (primary arm)

Expected Findings

Based on compound-signal literature: mSignals should detect sepsis onset **60–90 minutes earlier** than single-domain approaches at equivalent alarm burden. Specifically, the VAS/LRA divergence pattern (hemodynamically stable + lactate climbing) should predict sepsis-3 criteria with AUROC > 0.84 at 6h. AKI three-agent convergence (PHA nephrotoxin + VAS hemodynamic + LRA creatinine) should outperform individual-domain models by > 15% AUROC.

Publication Target and Strategic Value

Target journal: *Critical Care Medicine* or *JAMIA* — both accept MIMIC-based computational methodology papers prior to live clinical validation.

The MIMIC validation study serves three simultaneous purposes: 1. Evidence that the conflict-aware synthesis thesis is correct 2. The enterprise sales tool for hospital procurement conversations 3. The predicate evidence for FDA 510(k) submission

It is the single highest-leverage artifact in the mSignals commercialization path.

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11. Regulatory Pathway

FDA Software as Medical Device (SaMD)

mSignals operates as a Class II SaMD. The Clinical Decision Support provisions of the 21st Century Cures Act (Section 3060) provide a statutory exemption for software that:

1. Is not intended to replace clinical judgment
2. Displays the basis for each recommendation so a clinician can independently review it
3. Does not primarily acquire, process, or analyze medical images or patient-specific genomic data

mSignals satisfies conditions 1 and 2 structurally — the Human Gate model and conflict registry are designed to make the basis for every recommendation independently reviewable at the agent level.

For autonomous Tier-4 actions (rapid response paging), a **510(k) submission** with predicate devices — Tandem Control-IQ (autonomous actuation with governor threshold), Philips Guardian (ICU early warning auto-escalation) — is the clearance path. MIMIC-IV retrospective validation plus observational pilot data (Phase 4) constitute the clinical evidence.

HIPAA

All PHI is encrypted in transit (TLS 1.3) and at rest (AES-256), retained within the customer's BAA-signed environment, and logged to the immutable audit trail satisfying 45 CFR §164.312(b). The audit log produced by the Governor/Coordinator layer satisfies HIPAA requirements as a structural artifact, not an add-on compliance layer.

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12. Market Opportunity

Primary Market — US ICU

Segment	Size	mSignals Angle
US ICU beds	~55,000	Per-bed SaaS licensing
Top 20 Integrated Delivery Networks	Control ~35% of ICU capacity	Enterprise contracts
Annual preventable sepsis cost (US)	~$24B	ROI case for procurement
ICU monitoring market (global)	$3.5B, 8% CAGR	Platform displacement

Unit economics: A 400-bed hospital with 40 ICU beds. A 15% reduction in late-detected deterioration (conservative, within published compound-signal literature) generates $2–4M in annual downstream savings from reduced ICU length-of-stay, malpractice exposure reduction, and avoided readmissions. At $200–350K/year platform licensing, the ROI case closes in under 12 months.

Buyer Profile

Primary: Hospital CIO / CMO at an IDN-level health system. Decision drivers: Joint Commission alarm fatigue National Patient Safety Goal, CMS value-based purchasing penalties for sepsis mortality, malpractice exposure reduction, Epic contract renewal cycles.

Secondary: ICU medical directors seeking clinical tools their EHR vendor has not provided. Physician champion → department budget → system-level deployment.

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13. Extension Path: One Platform, Three Markets

The mSignals agent architecture generalizes across care settings by changing only the data source connectors and Governor Contract thresholds. The coordination logic is invariant.

```
graph LR
  subgraph P1["Phase 1 · ICU"]
    ICU["VAS + LRA + PHA\n+ NLA + IMA + HIA\nHigh frequency\nAcute care"]
  end

  subgraph P2["Phase 2 · Chronic Disease"]
    CDM["CGM + Activity + Diet\n+ Medication + HbA1c\nMedium frequency\nAmbulatory / home care"]
  end

  subgraph P3["Phase 3 · Behavioral Health"]
    BH["Voice + Sleep + Social\n+ Self-report + Wearable\nLow frequency\nMental health / outpatient"]
  end

  subgraph P4["Phase 4 · Population"]
    POP["Individual agents\naggregate to cohort\nPayer / public health"]
  end

  CORE["mSignals\nCore Platform\n(Governor + Coordinator\n+ Audit Layer)"]

  CORE --> P1
  CORE --> P2
  CORE --> P3
  CORE --> P4

  style CORE fill:#0a0a1a,stroke:#00f3ff,stroke-width:2px
  style P1 fill:#001a0a,stroke:#00ff9d
  style P2 fill:#0a1a1a,stroke:#00f3ff
  style P3 fill:#1a0a1a,stroke:#bd00ff
  style P4 fill:#1a1400,stroke:#ffd000
```

The same Governor Contract pattern that enforces a 30-second TTL on ICU vitals enforces a 15-minute TTL on CGM readings in a diabetes management context. The same conflict-aware synthesis that detects VAS/LRA disagreement detects sleep-agent/self-report disagreement in a behavioral health context. The platform is invariant; the vertical is configuration.

14. Conclusion

ICU patient deterioration is not a data availability problem. Every major hospital system has more clinical data than its clinicians can synthesize in real time. It is a data alignment problem: signals from independent systems, operating at different frequencies, describing the same patient, with no coordination layer between them.

mSignals provides that coordination layer. By assigning independent agents to each clinical data domain, enforcing Governor Contracts on each agent's confidence and freshness requirements, and synthesizing aligned patient states where inter-agent conflicts are first-class outputs rather than averaged-away noise, mSignals detects the compound deterioration patterns that no existing system can surface.

The architectural insight is not new. The artificial pancreas proved multi-agent patient state alignment works in a constrained domain. mSignals extends that architecture to the full ICU environment — six data domains, heterogeneous freshness profiles, conflict-aware coordination, and an audit-complete human gate layer designed to satisfy the regulatory requirements that keep single-model systems from reaching their full clinical potential.

The entry point is research-first: MIMIC-IV retrospective validation, a published study, a demonstration dashboard. That sequence proves the thesis on real ICU data with zero regulatory overhead and produces the credibility artifact that opens hospital procurement conversations. The path from there to FDA-cleared clinical deployment is long by design — that length is the barrier to entry that protects what comes next.

The signal is already there. mSignals aligns it.

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mSignals is a vertical of the ALI Multi-Agent Signals (MAS) platform. Build plan: `msignals-build-plan.md`*. Platform architecture:* `mas/implementation_plan.md`*. Related verticals:* `verticals/csignals/` *(chemicals),* `verticals/psignals/` *(payment rails).*